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Struggles Over Crop Rights and Labour Within Contract Farming Households in a Gambian Irrigated Rice Project

Judith A. Carney*

Contract farming schemes based on smallholders are expanding in Africa. Peasant household production is being restructured by the process. Research carried out in the Jahaly Pacharr irrigated rice scheme in The Gambia examines the changes that developed with the genesis of contract farming of the dietary staple, rice. The production routine generated conflicts and struggles within project households over access to and control of female labor. These have led to new labor processes in the project area, which are shaping producers' abilities to comply with contract farming production strictures.

I. INTRODUCTION

During the past decade there has been an expansion of agricultural projects in Africa based on contract farming production relations.¹ Watts [1986, 4–5] estimates that more than 60 contract farming schemes involving 16 commodities are currently operating in Africa. Although contract farming occurs in a variety of institutional contexts, it is held to offer considerable potential for smallholders. By making technological inputs available to those who would not otherwise be able to invest in them, it offers a means to overcome stagnation in smallholder production [Glover, 1983]. The new labour process imposed by contract farming, however, requires a restructuring of household production [Buch-Hansen and Marcussen, 1982]. While this effect has been described, there is little empirical research on how households adjust to these changes and its meaning for production.

This article seeks to address that gap by examining contract farming in a recently-implemented irrigation scheme in The Gambia. An analysis of the Jahaly Pacharr project offers an unusual case study of contract farming because the scheme was developed to boost productivity of the primary food staple, rice, and to achieve an equity objective. Women – The Gambia's traditional rice cultivators – were the intended beneficiaries. Commoditisation

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of the food crop has introduced significant changes in the property and family relations traditionally governing cereal production, particularly in the division of labour, household production and reproduction, land and crop rights, access to and allocation of household resources. These changes were conflictual and engendered struggles between men and women over access to and control of female labour. The household's ability to comply with contract farming production strictures has been shaped by the outcome of these intra-household struggles. The object of this article is to chart the main conflicts, struggles and changes that developed in project households with contract farming and to relate the forms of conflict resolution to currently-observed household production dynamics.

Observations on changing relations in household agricultural production were made during a seven month period in 1984 and from December 1986 to January 1987.² The Jahaly Pacharr irrigated rice project is located in central Gambia near Sapu, 280 kilometres east of the country's capital, Banjul. Irrigated rice production got under way with multilateral funding in the 1984 dry season.³ When completed in late 1987, the project will involve over 2,000 rural households from 70 villages. Nearly 1,500 hectares are developed in the project, of which 560 are pump-irrigated, some 700 tidal-irrigated (of which about one-third can be double-cropped), and 200 hectares improved for rainfed cultivation. The project directly involves four major ethnic groups: the Mandinka, Gambia's ethnic majority and primary rice cultivators, as well as the Wolof, Serrahuli and Fula.

II. THE JAHALY PACHARR PROJECT: THE HISTORICAL CONTEXT

The primary objective of the Jahaly Pacharr project is to increase production of The Gambia's dietary staple, rice, and thereby reduce foreign exchange costs currently experienced by dependence on imports for half the nation's annual needs. Despite the reliance on imports, rice is an indigenous food crop in The Gambia. The Senegambian region is one of the centres of domestication of *Oryza glaberrima*, which has been cultivated for at least a millenium [Porteres, 1970; Harlan et al., 1976; Carpenter, 1978]. The earliest European observers commented on its widespread cultivation [Jobson, 1623; Moore, 1738: 31] as well as the gender basis to rice production, noting it was grown by women [Moore, 1738; Park, 1799].

The accumulated experiences of earlier Gambian irrigation projects provide the context for understanding the social and economic objectives of the Jahaly Pacharr scheme. During the period from 1966 to 1980, pump-irrigated rice projects were successfully implemented in The Gambia. Nearly 2,400 hectares of swamp rice land were converted to irrigation schemes for the production of two crops per year.⁴ The schemes led to the successful introduction of 'Green Revolution' agricultural strategies to peasant producers but failed to achieve two fundamental objectives: (i) they were seldom double-cropped; and consequently, (ii) marketable surpluses were negligible. Furthermore, by 1980 less than one-third remained in cultivation. While the projects undoubtedly

contributed to the subsistence security of smallholders, they did not realise domestic rice import-substitution goals, and Gambian rice imports during this period climbed from 8,000 to 33,000 tons per annum [Carney, 1986].

While technical factors, such as faulty drainage designs and diesel and spare parts shortages, contributed to the problems of the small-scale irrigation schemes, social factors emerged as central. The irrigated projects operated under a similar development approach. The improved land and inputs were channelled to the male household head on the assumption that the household is a homogeneous production unit in which members pool resources and share in the rewards and benefits from marketable surpluses. This development approach had two important consequences. First, the arrangement did not facilitate double-cropping objectives. The household head only had the traditional right to uncompensated family labour for one cropping season. While household members would often join in dry season cultivation, during the rains male labour was diverted to the more remunerative groundnut crop. If the household head wished to cultivate a second crop, labour had to be remunerated. As a result, only about ten per cent of the developed plots were planted in the wet season. Second, awarding males the irrigated plots established the basis for their control of developed rice land [Dey, 1981]. The first phase of Gambian irrigated rice projects closed with the recognition that: (1) this approach did not release the family labour needed for double-cropping; and (2) under the customary tenure system, the government had no means to alienate un- or underutilised plots from smallholder producers.

Jahaly Pacharr was designed specifically to address these objectives. When completed in mid-1987, the project is expected to generate 6,000 tons of milled rice, about one-eighth of current imports. To ensure compliance with production targets, the Jahaly Pacharr project implemented contract farming. For households to remain in the project two fundamental criteria must be met: (1) double-cropping; and (2) the repayment in full of the seasonal production loan. While a written contract does not bind farmers to project guidelines, the threat of eviction does. Jahaly Pacharr was developed on lease land, which gives the government the right to remove plot usufruct from unproductive farmers.

Production decisions are centralised in the Jahaly Pacharr project. The management establishes the cultivation calendar for planting, weeding, transplanting and harvesting as well as specifying the seed varieties, types of fertilisers and pesticides used. Smallholders are informed of these decisions through the project's extension agents, who provide technical assistance. The project's paddy is marketed through the government's cooperatives.

By making inputs available to smallholders through a seasonal credit, the Jahaly Pacharr project is able to reach more farmers than previous 'Green Revolution' agricultural strategies, which favoured farmers with the resources to purchase the inputs. The project's assumption, thus, is that growth and productivity increases can be distributed in a more egalitarian manner. However, unlike the previous irrigation schemes, contract cropping requires a change in the organisation of household production, specifically

an intensification of family labour. Given the project's labour-intensive cropping demands, the Jahaly Pacharr project was predicated on the availability of family labour. Since project inception female labour, in particular, has been recognised as crucial to production objectives: '... women are better than men as far as transplanting is concerned and they are also better than men as far as working in the water ... so quite frankly we expect a lot of labour from women, more so than from men'.⁵

Female rice farmers were also the focus of the donors' interest in the project. Responding to a growing international critique of the adverse consequences for women of development interventions, and in particular the failure of previous Gambian irrigation schemes to reach women, the donors viewed the Jahaly Pacharr project as an opportunity to implement equity objectives: 'In assisting the Government [of The Gambia] to reach its goal of rice self-sufficiency and to improve the lot of the rural poor, the project makes special reference to women, who traditionally have been the major rice growers under arduous swamp conditions' [IFAD, 1981]. The donors' intent was not only to integrate women into the project but to make females the primary beneficiaries by awarding them the developed land [IFAD, 1984].

Gambian Farming System: Property and Labour Relations, Crop Rights and Agrarian Change

In order to understand the conflicts that emerged with contract farming in Jahaly Pacharr households, this section provides a background to the main factors traditionally conditioning production dynamics. Two fundamental features characterise the Gambian farming system. First, cropping patterns have evolved out of response to a short, five-month growing season and, second, the principles governing resource use and access have been adjusted to labour, rather than land, shortages. While this discussion is focused on the Mandinka – the dominant ethnic group and primary rice cultivators in the project area – the basic features of the farming system are similar across the production systems of the other participant ethnic groups.

The Gambian farming system operates simultaneously with several types of tenure relations, which carry specific crop rights and labour obligations. The most widespread form is land held in customary tenure by the household [compound] residence unit, which is known as *maruo*. This land provides the material basis for household reproduction and cannot be alienated from the kin-residence group. Besides referring to all the fields owned by the household, *maruo* also refers to a set of labour obligations. As members of a farm unit, all able family members are traditionally expected to provide labour for household subsistence. Men's *maruo* obligations are traditionally met on the uplands where they cultivate millet, sorghum and maize or grow groundnuts to purchase cereals. In rice-growing areas, women cultivate it as their *maruo* crop. Produced for the household's subsistence needs, *maruo* crops come under the control of the male compound head or other senior males, who arrange its storage and distribution.

In the Gambian farming system a second, and important, sub-set of tenure relation operates on *maruo* land. In exchange for providing labour towards household subsistence, dependent males and females are given the right to cultivate some *maruo* land for their own needs. These usufructuary land rights are distinguished linguistically and known as *kamanyango*. As long as the cultivator remains a member of the household, she/he controls the plot's use and rights to the crop produced. *Kamanyango* usufructuary rights are an important means through which dependants obtain cash from farming. Unlike the *maruo* plots, the cultivator cannot make any claims to family labour although sometimes she/he receives voluntary assistance from relatives and friends. Among the Serrahuli, Fula and Wolof ethnic groups *kamanyango* plots are usually located on the uplands and cultivated in groundnuts. However, when conditions permit Mandinka women grow rice as both a food and cash crop.

Finally, individually-owned land also exists in The Gambia. It, too, is known as *kamanyango*. Individual claims are established through the act of clearing unclaimed land. In rural Gambia, individually-owned *kamanyango* occurs primarily in rice zones. Women's individual claims have been made in tidal and rainfed rice areas but not on pump-irrigated plots, where men have established control. Individual *kamanyango*, like usufructuary *kamanyango* on household *maruo* plots, does not enable the cultivator to draw on the labour obligations of family members. When supplemental labour is needed women utilise their informal, reciprocal labour networks while men frequently hire help on their individual plots. Unlike household-owned land, individual land can be alienated as the cultivator wishes. For women this has been an important way to alienate rice plots to daughters who move into resource-poor households [Dey, 1980]. Despite the existence of individual land ownership in rural Gambia, strong sanctions operate against its sale. Irrigated rice plots, however, are often rented.

Gambian rice development projects have been accompanied by conflicts over women's land and crop rights. These have primarily affected Mandinka women, who are specialised rice producers. The first documented conflict surfaced in the late 1940s when the colonial government constructed causeways and footbridges to improve access to the tidal swamps. Women began clearing the areas for cultivation, which men contested because it established the basis for women's land ownership. In one case that was brought to the attention of colonial authorities, men set forth their reasons why women should not own land: '... if women mark the land and divide it, it will become "women's property", so that when the husband dies or divorces his wife, the wife will still retain the land, which is wrong. Women must not own land.'⁶ While the quote suggests that there may have been a precedent to women's ownership claims to swamp land, rice development was clearly causing these rights to be reinterpreted. Men's resistance to women's land ownership centered on the issue of divorce. Since residence in Gambian villages is virilocal and divorce common, the plots would be alienated from household control when marriages dissolved. This legitimised men's claims to the colonial authorities who

subdivided the cleared area between male compound heads, thereby making the land household, rather than women's individual, plots [Rahman, 1949].

The same type of conflict over women's rights to rice land has accompanied development projects during the independence period. In the 1950s, for instance, women became tenant rice farmers in a Colonial Development Corporation (CDC) project in the present Jahaly Pacharr area. When the CDC project failed and women tried to assert their *kamanyango* rights to the plots, men claimed the area as *maruo* [Dey, 1980: 252–3]. Non-governmental organisations working in Gambian tidal rice development have likewise acquiesced to male pressure to designate the areas as household rather than women's individual fields.⁷ As in the colonial period, men have defended their position on the grounds that divorce would enable women's *kamanyango* to circulate out of the household. But the significance of the *maruo* designation for resource struggles is that when it has been applied, females have lost their *kamanyango* crop rights. The renaming of the developed rice plots has thus functioned to erode women's individual crop rights and to bring female labour power and the products from it, increasingly under control of the household head.

In sum, Gambian rice development has been accompanied by resource struggles between men and women. These have been waged and won by manipulating language to redefine women's customary land and crop rights within the farming system. With the application of *maruo* to developed rice land, a new tradition has developed within the farming system which operates to deny women's rights to rice land and, importantly, their control over the products of their labour. Irrigated rice development, moreover, has enabled men to establish direct control over the plots either through individual ownership or by designating the plots as household fields and thereby, drawing on females' *maruo* labour obligations to cultivate them. Yet the first phase of Gambian irrigated rice development remained adjusted to practices that had long characterised the farming system. Even though naming the irrigated plots *maruo* acted to extend women's agricultural calendar into dry season cultivation, they were not obliged to provide unremunerated labour for two cropping periods. During the rains females continued to exercise their traditional right to cultivate *kamanyango* crops. This was possible because they still had access to rice land.

III. CONTRACT FARMING AND GENDER RESOURCE STRUGGLES

The first land distribution set the stage for the conflicts and struggles that developed in Jahaly Pacharr households over the project's equity and productivity goals. Even though the donors had specified that the original female rice tillers were to receive the irrigated plots, the first distribution proved this would not be easily achieved. Less than one-third of the plots were registered in women's names. Most of the plots, instead, were allocated to male household heads. The results generated conflicts on two fundamental levels: (1) among men and women; and (2) between the project management and the donors.

The new labour process implemented in the Jahaly Pacharr project intensified the ongoing conflicts between males and females over women's rights to land. Since participation in the project depended on compliance with contract cropping schedules, skilled female labour became an important asset. The development of the project, however, placed Mandinka women in a particularly vulnerable position. As the only ethnic group in which women have no upland *kamanyango*, the absorption of the pre-existing swamp rice fields into the project meant that women's ability to cultivate a plot for cash would henceforth largely depend on access to irrigated plots. Unless their rights were protected, Mandinka women had no alternative cash-cropping areas and importantly, no protection from household claims to their labour and the products from it. If blocked from access to land, women were well aware that they would lose a fundamental right they had traditionally enjoyed within the farming system – the right to an individual field in exchange for the labour they provided on household crops. Their vulnerable position was forcefully presented by one Mandinka woman: 'It seems this project is just like the Chinese one when we suffered before. We aren't going to put up with that again ... I have this to say to you men. We women aren't going to accept the way we have been treated in the past. We were asleep then. But now we are awake.'⁸ But, as the land distribution demonstrated, men resisted plot awards to women. The conflicts that had characterised earlier rice development projects surfaced once again in Jahaly Pacharr as men justified their opposition to female land control on the grounds that with divorce, the plot would be alienated from household control.

The other source of conflict in the land distribution was between the donors and the project management, which was charged with implementing the equity objectives. Compliance with the donors' mandate was constrained by political and economic considerations. In the absence of a land nationalisation law, the Gambian government had to negotiate the project's lease with local farmers. To do so, officials mobilised the support of the area's traditional elites who secured the backing of male household heads. While a lease was negotiated, it was made subject to renewal after 21 years. Thus, long-term project viability depends on maintaining the support of these political linkages with men who were opposed to major changes in resource control. The government's interest in equity goals, furthermore, was secondary to its concern with productivity. Because the project's production routine is predicated on an intensification of family labour, government officials were unwilling to champion principles that might reduce its availability. As a labour category, *maruo* alone provided the institutional base to make claims to family labour. Since individual *kamanyango* fields do not benefit from unremunerated family labour, the project management was reluctant to risk implementing a major resource change that would award individual women the land.

The failure of the first land distribution to follow the project's primary funding objective caused the main donor, IFAD (International Fund for Agricultural Development), to intervene. A mission was sent to The Gambia in early 1984, which succeeded in re-registering most of the project's plots in

women's names. IFAD's action received widespread international attention as evidence for the achievement of gender equity in rural development [*New African*, 1985; *International Agricultural Development*, 1985; *African Business*, 1986], but the emphasis on the land registration obfuscated the real changes that had occurred in control over the project's land.

IFAD was able to re-register the plots in women's names because male heads of household had already achieved *de facto* control. The project management had successfully overcome male resistance to IFAD's land registration by agreeing that the household, not the individual woman, had final control over the project's land [IFAD, 1984]. Given the prevailing virilocal residence pattern this seemed reasonable to the donors, who were assured that women's *kamanyango* crop rights would be upheld. The management's decision, however, cleared the way for men to declare the project's plots *maruo* – the issue that was the real locus of gender resource struggles.

The importance of language in expressing control over land and specifically, female labour, soon became evident. During the first year, all the pumped plots were termed *maruo*. Women provided most of the agricultural labour, but household heads refused to give them the promised *kamanyango* crop rights. During the second cropping season when it became obvious that they were not going to receive any benefits from their labour on the pumped plots women began to press for official acknowledgement that the tidal and rainfed areas under development would become their *kamanyango*. Despite the management's earlier avowal of women's individual crop rights on project land, it did not intercede to provide the necessary official support to ensure that possibility. Nor did IFAD intervene at this critical juncture to safeguard women's rights in the project. As a consequence, in order to secure *kamanyango* crop rights in the tidal and rainfed areas, women were forced to negotiate through the very village and household structures which sought to deny them *kamanyango*.

The differential effects of project development on household members were clearly discernible during the first year. Women provided the bulk of the field labour, while the compound head maintained his traditional control over *maruo* production. However, the transformation of rice in the project area from a subsistence to a cash crop, and thus from a use to an exchange value, renders the traditional sanction against selling the product of *maruo* meaningless. From the project's first harvest it has been the male household head who markets and sells the plot's rice. And it is he who keeps the revenues from paddy sales. The meaning of this for resource control and accumulation is not insignificant. Yields on the project's pumped plots have averaged over 5.6 tons per hectare while the sale of paddy to repay the seasonal credit has been fixed at about 1.2 tons, less than half the average annual yield. Given the low pre-project average per capita income (US\$130) and the value of one ton of paddy (1986: US\$124) the surplus represents a substantial income increase. The naming of the project's plots, *maruo*, thus enables the compound head to make claims to women's unpaid labour when in fact the plot functions as his individual *kamanyango*.

This use of '*maruo*' thus both embraces and deforms the term's original meaning. It has enabled the household head to establish control over the project's plots and therefore, to female labour. Rather than broaden women's crop rights and land access, development of the Jahaly Pacharr project has accelerated what Berry [1986] terms 'concentration without privatisation' and in the process has greatly increased the power and accumulation possibilities of senior household males. But women have not passively accepted such changes.

IV. PROJECT IMPACT ON INTRA-HOUSEHOLD PRODUCTION DYNAMICS

This section examines the influence of contract cropping on the labor process within Jahaly Pacharr households. The emphasis is on women's responses to their husbands' claims over their labour on project plots. Here the intent is to document the forms of gender struggles in household production and how they are being resolved, in order to illuminate their relevance for productivity and output and thus project success. The importance of female labour to the household's ability to follow basic production goals presented what Scott [1985] terms a 'weapon of the weak' – specifically, the occasion for women to challenge their husbands' claims to their labour.

By the end of the project's first cropping season, women had received no *kamanyango* rights in the project. Nor had they been remunerated in cash or kind for their labour on the pumped plots. The importance of their labour to retention of plot usufruct, however, provided the basis for women's struggles to improve their position within the household. At the onset of the project's second year, women began to demand from their husbands remuneration for their work on the pump-irrigated plots.⁹ In so doing they are contesting the basis of the conjugal contract, namely the right of their husbands to control over their labour power¹⁰ [Whitehead, 1981; Jones, 1986], and thereby effecting from within the household a challenge to patriarchal family relations [Friedmann, 1986].

Women's success in bargaining for labour remuneration has been conditioned by three primary factors: (1) their ethnic group; (2) the economic position of their household; and (3) conjugal relations. The most determinant factor in women's bargaining position was whether or not they still had access to individual fields. Project development did not change the land and crop rights of Wolof and Serrahuli women, who retain upland groundnut *kamanyango*. They maintain the right, like junior males, to withdraw their labour from household demands during the rainy season – a consideration which enhanced their intra-household bargaining position. As a result, in households with adequate project land to generate marketable surpluses, these women generally receive a portion of the plot's paddy as labour remuneration.¹¹

The pattern is more complex and contradictory among the Mandinka. Most Mandinka women lost their *kamanyango* rice fields with project development. The *maruo* designation left them without alternative land access, thus placing

them in a vulnerable position to their husbands' efforts to 'capture' their labour for rice production. Expectations for Mandinka women to perform most of the cropping operations were not without historical precedent. Since the nineteenth century the Mandinka have responded to economic commoditisation by intensifying the spatial and gender basis of production, specifically by increasing food cropping in the tidal swamps and female labor in rice production [Weil, 1973; Carney, 1986]. The implementation of contract cropping in Jahaly Pacharr required an intensification of family labour, thus rupturing the gender basis of Mandinka rice production. The struggles that ensued in Mandinka households were over the degree to which female family labour would bear the work burden. Mandinka women's weakened negotiating position with loss of land access has had one important consequence. Even though the Mandinka received more land than other ethnic groups, due to their dominance in pre-project rice cultivation, women have not always been able to negotiate labour remuneration.

By the end of the third cropping year three major responses could be identified by Mandinka women to the changes that had occurred in household property and family relations:

- (1) Women provide labour as needed on pump-irrigated plots and receive compensation through the usufruct of the project's tidal-irrigated and rainfed plots.
- (2) Women provide labour as needed on the pumped plots, but since no tidal areas are available for cultivation, they are compensated in paddy, receiving a fixed share of the pumped plot's yield for their labour.
- (3) Women do not provide labour on the household *maruo*. The household head cannot, or refused to, offer labour remuneration in paddy on the pumped plot or through access to non-pumped plots.

In the first adaptation, the households involved were usually resource-sufficient, controlling adequate project land for both subsistence and cash needs. Given that nearly 70 villages are involved in the Jahaly Pacharr project and the fact that tidal land was still being developed in January 1987, it is difficult to estimate the overall percentage of Mandinka households in the first category. In one village intensively surveyed, Pacharr, which received land in both the pumped and non-pumped areas, women have received *kamanyango* rights to the tidal-irrigated plots. However, their economic opportunities in the project have been tempered by four considerations. First, most plots are not cultivated by one woman. Each one hectare plot is often shared between as many as four females. Second, in contrast to the pumped plots, the majority of tidal, and all the rainfed, areas are cropped for only one season which limits the plot's cash-generating possibilities. Third, these plots tend to be less productive than the pump-irrigated areas. Finally, a woman's *kamanyango* crop rights are only protected while she remains in the household. With divorce, she can lose them.¹² Despite such considerations, *kamanyango* usufruct to tidal and rainfed plots do provide Mandinka women some

form of labour compensation and thus, cash-earning possibilities in the project.

The second adaptation to Mandinka women's 'wage' demands occurred in households which had ample pump-irrigated land but little tidal or rainfed land. Women have sometimes successfully negotiated elementary share-cropping relations, an adaptation which rewards them for their overall labor productivity. In exchange for their unrestricted labour on the pumped fields, women are given a fixed percentage of the plot's yield. The percentage of households involved in sharecropping arrangements in one Mandinka village intensively surveyed, Wellingara, was about 20 per cent. The yield percentage women received varied but averaged about ten per cent, with a paddy value that ranged between 150 and 500 dalasis per year (US\$20-\$70).

Most of Wellingara's households, however, fell within the third category. Women were unsuccessful in receiving labour compensation. In a few cases this was due to the household's economic position. About 15 per cent of Wellingara's households did not receive much project land. The plot is shared with other households and functions primarily for subsistence rather than generation of cash income. These circumstances constrain women's ability to overcome their *kamanyango* loss which has led them to pursue other economic activities by hiring themselves out as wage labourers, growing market gardens, and/or engaging in petty trade.

The significance of the impact of changing intra-household resource control on Mandinka women is most clearly visible in the majority of Wellingara households which have enough land to provide for subsistence and cash needs, but refused to remunerate women for their work on the household's plots. As a consequence, women have withdrawn their labour from the rice fields. Female labour withdrawal has had a far-reaching impact on the social organisation of household production.

Many Mandinka households are having problems adhering to contract cropping production schedules, which has contributed to lower yields. In the project's first year, average dry season pump-irrigated rice yields were 7.5 tons per hectare. In the 1985 and 1986 dry seasons they have averaged 5.7 tons.¹³ While the yield declines do not put the Jahaly Pacharr project in jeopardy, they have seriously reduced anticipated productivities. The lower than average yields experienced by many of the project's Mandinka households is the basis for the management's oft-repeated remark that the Mandinka are the project's worst farmers.

Another consequence of female labour withdrawal has been an intensification of male labour in rice cultivation. But given the high rate of polygamous marriages in the project area, men cannot compensate for the loss of skilled female labour through an intensification of their own. Thus, there has been an increase in labour hire by these households, particularly in the rainy season when men also plant groundnuts. Even in the project's first year when women provided unpaid labour on the plots, only 25 per cent of Wellingara's households were able to complete transplanting without labor hire. By the project's third year when the effects of female labour withdrawal were in place,

more than half the village's households were also hiring labour for harvesting and threshing.¹⁴ For the 1986 cropping year such households paid between US\$30 and \$85 for hired labour, which closely approximated the value of paddy women are receiving in sharecropping arrangements. Female labour withdrawal has thus affected the capacity of the household to intensify its labor, a primary objective of the project.

Finally, most of the wage labourers in Mandinka villages are the females who have become proletarianised with project development. Their economic activity is now concentrated in providing wage labour on the project's plots, which has contributed to an erosion of women's reciprocal labour networks. Traditionally village women's age grades, known as *kafos*, provided large labour groups for rice transplanting. The money collected was used for common purposes such as financial assistance in time of need or for ceremonies. At other times these reciprocal networks functioned as an auxiliary labour reserve for women who were sick, childbearing or otherwise unable to go to the fields. Nowhere are the project's effects on the labour process more visible than in the transformation and individualisation of women's *kafos*. For females dispossessed of rice land and currently working for hire on the project's plots, the *kafos* now operate as work groups. They provide an organisational framework in which women pool their labour for hire in, transplanting, weeding, harvesting or threshing. In contrast to the pre-project period, the money collected is no longer retained for the group's mutual purposes but divided up among the individual female members. By forming work groups women are able to receive a better rural wage than as single wage workers. For example, a group of 20 women usually charge 80 dalasis for transplanting one irrigated plot and they can usually complete two a day. Each woman then makes eight dalasis instead of five [US\$0.70], the daily wage labour rate. Thus, for women who no longer have land access, their economic activities are increasingly met through wage labour on the project's rice plots, which has led to an individualisation of their reciprocal labour networks.

The outcome of resource struggles in Jahaly Pacharr is by no means settled. Project households are still negotiating women's rights to remuneration for their labour as well as the form and rate of their labour value. Though the resource position of households has been important in conflict resolution, more basic factors were: (1) whether or not women maintained *kamanyango* plots with project development; and (2) the degree to which male household heads were willing to compensate women for their land loss. The resolution of such conflicts is central to the project's productivity goals and to the viability of contract farming as a mechanism for effecting anticipated increases in smallholder productivity.

V. CONCLUSION

In the past few years, research on African agrarian change indicates that labour is frequently the critical limiting factor in agricultural production [Dey, 1981; Watts, 1983; Berry, 1986; Richards, 1986]. Smallholder-based contract

farming schemes, predicated upon its availability and capacity for intensification, entail fundamental adjustments in the organisation and division of household labour. Such changes, however, are mediated through patriarchal family relations and kin-age-gender-based institutional structures of labour mobilisation and resource access. The imposition of intensified commodity production in this setting becomes conflictual, generating intra-household struggles as the new labour process is internalised by the household. In contract rice farming households in The Gambia, the tension between commodity relations and family relations erupted most dramatically along gender lines, specifically over men's efforts to gain access to and control over their wives' labour power, and women's resistance to these claims.

In the project area it is frequently said that 'rice is a woman's sweat'. With the imposition of contract farming, women began to demand remuneration for their toil – either through usufruct rights to tidal and rainfed plots – or through sharecropping the pumped plots. When this was not forthcoming, they withdrew their labour from the household's plots. While women have been proletarianised as a result, they have also been liberated from their traditional burden in subsistence production, which men increasingly shoulder.

Contract farming of a food crop has thus engendered changes in property relations and family relations, transforming in the process production dynamics, social relations and the labour process. Such changes are still occurring as project households adjust to the production routine and women's demands for labour remuneration. While contract farming has improved the subsistence security of the majority of project households, its viability for achieving national rice import-substitution objectives depends on the implications of conflict resolution for household production.

NOTES

1. Contract farming is a form of production in which farmers supply agricultural commodities to a buyer/processor under conditions established by a written or oral agreement [Watts, 1986]. These contracts, rather than the farmer, shape production decisions and inputs. Often the seed varieties, fertilisers and insecticides used are specified and sometimes the cropping calendar and timing of field operations.
2. Interviewing techniques involved quantitative surveys as well as participant observation, key informant interviews and discussions with male and female farmers. The seven villages selected for in-depth research represented each project ethnic group (four Mandinka; one Fula, Serrahuli and Wolof); those with pump-irrigated plots only; and villages with plots in pump-irrigated and tidal or rainfed areas.
3. The Jahaly Pacharr project is funded with a US\$16.5 million loan on easy lending terms through the International Fund for Agricultural Development (IFAD), the African Development Fund, the World Food Program and the governments of the Netherlands and West Germany.
4. The aid missions involved in the first phase of Gambian irrigation were: Taiwan (1966–74), World Bank (1973–76), and Peoples Republic of China (1975–79).
5. Jahaly Pacharr project manager quoted in BBC's Global Report, 'The Lost Harvest', 1983, produced by Sarah Hobson.
6. From Notes on 'Land Tenure' in Genieri, The Gambia [Rahman, 1949], cited in J. Dey [1982].

7. The principal non-governmental organisation involved in tidal rice development is the German government-funded Freedom From Hunger Campaign which has improved access to about 5,000 hectares of swamps and distributed seeds and fertilisers.
8. Mandinka woman quoted in BBC's 'Lost Harvest'. Her reference is to the development approach pioneered by the Taiwanese Mission and followed in subsequent small-scale irrigated rice projects (note 4). The Taiwanese introduced pump-irrigated rice farming to male household heads, assuming that households operated with a joint utility function. But not all family members equally labored in the irrigated plots, nor were benefits from marketable surpluses distributed equitably. By targeting male household heads for technological diffusion, the missions provided a legitimization for men's control over pump-irrigated perimeters. Women suffered doubly: (1) they were excluded from the benefits deriving from technical improvements in rice production; and (2) since the irrigation projects were often developed in tidal rice areas, women lost many of their *kamanyango* plots.
9. Jones [1986] reports a similar finding in households involved in irrigated rice production in northern Cameroon.
10. The usage followed here is set forth by Whitehead [1981: 88] and refers to 'the terms on which husbands and wives, exchange goods, incomes, and services, including labour, within the household'.
11. The discussion refers principally to Serrahuli women who grew rice as a *maruo* crop and groundnuts as *kamanyango* prior to project development. Based on their previous rice cultivation in the Jahaly and Pacharr swamps, Serrahuli and Mandinka villages have received more project land than Wolof and Fula villages.
12. This is a major source of litigation in the project. The cases that have been adjudicated have established the following precedents: (i) if a woman divorces her husband, she loses the *kamanyango* crop rights; (ii) if he divorces her and she remains in the project area, she has the right to continue cultivation.

13. TABLE 1
SEASONAL YIELDS ON PROJECT PLOTS

Season	Year	Yield/hectare
Dry	1984	5 tons
Wet	1984	6.5
Dry	1985	5.7
Wet	1985	4.0*
Dry	1986	5.7
Wet	1986	4.8 (est.)

Note: * this decline reflects the lowered production of the tidal and rainfed plots which began being developed in the project's second year.

Source: Project Management Unit, Jahaly Pacharr Project.

14. TABLE 2
WELLINGARA: PERCENTAGE COMPOUNDS HIRING LABOUR ON PROJECT
MARUO

Activity	1984 ¹	1986 ²
Transplanting	67%	66%
Harvesting	42%	61%
Threshing	21%	63%

Notes: ¹ sample: 25 households [of 40]

² sample: 38 households [of 40]

Source: Fieldwork data.

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